Does weight decay bound the free recurrence?

exp063 · 11 July 2026 · Draft

Abstract

Weight decay regularises the free net but does not stabilise it. The Spiking Heidelberg Digits (SHD) program already has a stable recipe from exp062: imposing Dale's law (each neuron's outgoing synapses share a fixed sign, all excitatory or all inhibitory) trains the network NaN-free where the free signed net, whose weights may take either sign, diverges. But that constraint costs a few points of test accuracy (53.2% against the free net's 56.8%), the price of denying the signed recurrence its expressivity. So the live question is whether stability can be had **without** that tax: can a plain regulariser bound the free net's runaway dynamics instead of a hard constraint? To answer it, this sweeps the AdamW weight decay $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$ on the free net and watches the recurrent weight norm W_{ee} and the NaN-epoch rate. The verdict is that decay fails the test: it improves generalisation but never bounds W_{ee} or removes the numerical divergence.

Methods

1. Fix the free net (signed recurrence, no Dale's law) to the exp060 recipe at integration timestep $\Delta t = 1$ ms, the setting where it diverges, single seed, at full run scale.
2. Sweep the AdamW weight decay λ (a penalty that shrinks every weight toward zero at each step; λ sets its strength) over $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$. Weight decay is the **only** thing that varies, so any change in the stability metrics is attributable to λ alone.
3. For each λ, record the plan's pre-registered contrast: the maximum norm of the recurrent excitatory-to-excitatory weight matrix W_{ee} and the NaN-epoch rate (the fraction of training epochs whose loss or activity went non-finite), together with the best test accuracy.

A first pass at $\lambda = 10^{-3}$ tamed the gradient explosion but did not bound W_{ee} or remove the NaN, so the sweep reaches up to 10^{-1}.

> **Kill criterion** (the pre-registered condition that would rule decay out): if even the strongest decay leaves W_{ee} growing and NaN still present, decay is not part of the stability recipe. It regularises but does not stabilise.

Parameter	Value
Dataset	SHD, 1000-sample subset

Model	COBANet (conductance-based spiking network), 256 hidden units, all four recurrent weight blocks (W_{ee}, W_{ei}, W_{ie}, W_{ii}) trainable, signed (no Dale's law)
Swept variable	AdamW weight decay $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$
Integration	$\Delta t = 1$ ms, $T = 1000$ ms (the setting where the free net diverges)
Seeds	1 (seed 42), a quick sweep
Training	30 epochs, learning rate 0.001, batch size 32, surrogate-gradient backpropagation through time (BPTT)
Regulariser	upper firing-rate bound (threshold $\theta_u = 100$, strength $s_u = 0.06$)
Compute	runpod

Results

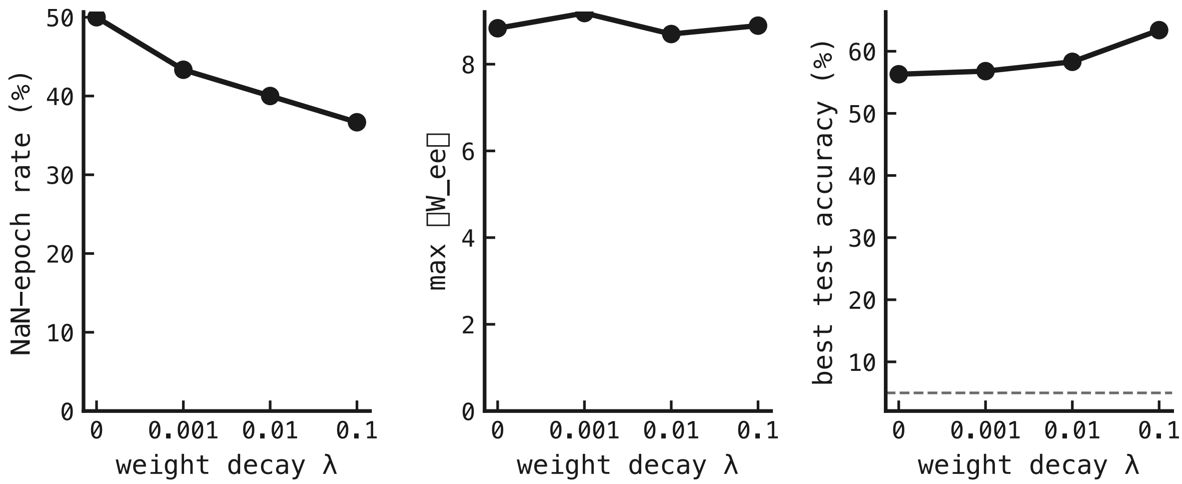

Figure 1: Three stability metrics against the AdamW weight decay λ (weak to strong), one per panel: (left) the NaN-epoch rate, the percentage of training epochs whose loss or activity went non-finite; (middle) the maximum norm of the recurrent excitatory-to-excitatory weight matrix W_{ee}; (right) the best test accuracy in percent, the dashed line marking chance (5%). **What we expect if decay stabilises:** the NaN-epoch rate and the runaway weight norm fall as λ grows, ideally reaching zero NaN while accuracy holds above the Dale's-law recipe. **What we see fails the test.** The NaN-epoch rate falls only a little as the decay strength λ grows, from 15 of 30 epochs at $\lambda = 0$ to 11 at $\lambda = 0.1$, and never reaches zero; the recurrent weight norm W_{ee} barely moves (≈ 8.8 throughout), so decay does not bound the runaway weight. One quantity does improve monotonically: best test accuracy climbs from 56.3% to 63.4%. Decay is a good regulariser (it improves generalisation), just not a stabiliser (it does not prevent the numerical divergence).

The kill criterion fires. Even the strongest decay ($\lambda = 0.1$) leaves the free net diverging (11 of 30 epochs go NaN) and the recurrent weight norm W_{ee} is essentially flat across the whole sweep (8.9 at the strongest λ against 8.8 at zero). Weight decay neither bounds the recurrent weight nor removes the divergence. It *regularises*: accuracy rises steadily with λ, from 56.3% to 63.4%, the free net's best number yet, but that is a generalisation effect, not a stability one. Decay is not part of the stability recipe.

The picture is now complete for the three soft knobs. A soft knob here is a training setting the free net can vary without changing the model equations. Of the three tested (the integration timestep Δt in exp061, Dale's law in exp062, and weight decay here), only Dale's law stabilises. Δt makes the gradient explosion *worse*; weight decay regularises without bounding W_{ee}; only the hard non-negativity constraint of Dale's law holds the excitatory-inhibitory-excitatory (E→I→E) feedback loop gain below runaway. So the free signed net's divergence is not reachable by any soft knob tested. Stabilising it *while keeping its accuracy* needs the reserved model change: a forward-pass state clamp that bounds voltage and conductance so a diverging trajectory cannot reach NaN.

A hint for that next step. Decay's accuracy gain (up to 63.4%, the free net's best here and well above the Dale's-law recipe) says the signed net has headroom that the constraint gives up. A state clamp that stabilises the free net *at strong decay* could plausibly beat every recipe in the program so far. That is the experiment the plan now points to.

Caveat. Single seed; the effect is categorical (NaN present at every λ), so it is unlikely to reverse, but a 3-seed confirmation is cheap to run.